

Your grade: 100%

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Next item →

1. When prompting an LLM to generate a query for your database, which of the following would it help to include in your prompt to improve the quality of the results. Select all that apply.

1 / 1 point

☒ Assign a role to the LLM, for example a database querying expert

✔ **Correct**
Assigning a relevant role can often lead to higher quality results

☒ Include the schema of your database

✔ **Correct**
Providing the schema of your database will help the LLM write queries that will correctly use it

☒ Include the library you will use for querying, for example SQLAlchemy

✔ **Correct**
Details like this can help an LLM use the correct libraries when writing your queries

☐ The contents of your database in an easily shareable format like CSV or JSON

2. Which of the following best describes the benefits of indexing columns in a database?

1 / 1 point

- ☐ Indexing generally cannot improve the performance of an application.
- ☐ Indexing additional columns will always result in greater application performance
- ☐ Column indexing decisions made by LLMs can be trusted without testing
- ☒ Indexing, when done sparingly and with attention to properties of the column data, can result in better performing applications

✔ **Correct**
Indexing can often yield performance boosts if done sparingly and with attention to the properties of the columns being indexed.

3. Which of the following best describes query caching?

1 / 1 point

- ☐ Query caching stores all SQL queries made on a database to help identify the most common queries that are called
- ☒ Query caching stores the results of queries, potentially saving resources that would be used to run commonly-called but expensive queries
- ☐ Query caching saves the state of the database at different points in time so that the same queries can be run on historical versions of your data
- ☐ Query caching stores smaller chunks of your database in order to speed up expensive queries by running them on a smaller amount of data

✔ **Correct**
Query caching stores the results of queries so that they can be returned when that query is received again without actually performing the query in your database. Especially with commonly-called an expensive queries, this practice can save resources and improve application performance.

4. When debugging your database, which processes is an LLM able to support you with?

1 / 1 point

☒ Translating the results of an EXPLAIN query to a more human-readable format to help you interpret result

✔ **Correct**
As shown in the lecture, LLMs can support you with this process

☐ Running test queries on your database for you to identify unexpected behavior

☒ Brainstorm potential libraries and commands that can be used to brainstorm various issues

✔ **Correct**
As in many programming contexts, brainstorming potential solutions to problems is an area where LLMs are very capable.

☒ Rewriting parts of your code to implement useful debugging features like logging

✔ **Correct**
LLMs will be able to help rewrite your code to implement useful features like logging at your direction.

5. What debugging process is the following code designed to implement?

1 / 1 point

```
1 def explain_query(query):
2     result = session.execute(text(f"EXPLAIN {query}"))
3     return result.fetchall()
4
5 query = "SELECT * FROM users WHERE id = 1"
6 explain_result = explain_query(query)
7 print(explain_result)
```

- ☐ Logging queries to a file to be reviewed later
- ☒ List the order in which a query accesses tables
- ☐ Identify records in a table that make duplicate use of the same unique id
- ☐ Cache the results of the provided query to be accessed later

✔ **Correct**
The EXPLAIN command does exactly this, helping reveal the execution plan of your query.